

Team Name: Falcons

Track: Predictive Analytics (ML / DL)

Problem Statement:

Develop an AI-powered automated evaluation system for handwritten answer sheets. Instead of manually programmed marking rules, the platform **learns how teachers grade** from previously corrected answer scripts, then automatically evaluates new scanned scripts, assigning **question-wise marks, totals, percentages, and feedback** with human-like consistency, fairness, and transparency.

Sub Part 1 (Dataset - How it is being implemented):

A **structured training dataset** (CSV / SQL): every historical student answer is paired with its official answer key and the **teacher-awarded mark (the label)**, from past corrected scripts, question papers, model answer keys, and rubrics.

Sub Part 2 (Training Engine - How it is being implemented):

A **custom-trained mark-predictor** learns teacher marking behaviour, not hand-written rules. Engineered features (**semantic similarity**, concept coverage, keyword recall, missing/extra points, negation cues) feed a **Gradient Boosting regressor (XGBoost / Random Forest, neural-net option)**.

Sub Part 3 (Evaluation - How it is being implemented):

Measured on a held-out split of teacher-marked data: the system reports **RMSE, MAE, R^2 , and accuracy within +/-1 mark** of the teacher's marks, plus feature importance, quantifying how closely it reproduces real teacher grading.

Sub Part 4 (Output - How it is being implemented):

For a new script: **handwriting OCR** then question segmentation and answer-key matching; the trained model predicts **percentage-based marks** per question. Output is a fully evaluated sheet: marks, total, percentage, feedback, and **deduction reasons**. Unreadable answers are flagged for human check, never guessed. **Bonus:** feature engineering + hyperparameter tuning.

Why ExamShield Beats ChatGPT & Generic AI Evaluators

Dimension	ChatGPT / General LLMs	CBSE-style / Cloud AI Evaluator	ExamShield (Ours)
How the mark is decided	LLM text generation - guesses , can hallucinate a score	Central black-box cloud model	Trained ML model on real teacher marks - a deterministic prediction
Learns YOUR teachers' marking	No - generic knowledge	Not institution-specific	Yes - trained on your own corrected scripts
Explainability	Opaque; no reliable reasons	Limited transparency	Per-answer feedback + deduction reasons + feature importance
Measurable accuracy vs teachers	Not reported	Rarely disclosed	RMSE / MAE / R^2 / +/-1-mark accuracy
Consistency	Varies per prompt / run	Centralized but not tuned to you	Same learned standard for every script
Data privacy	Scripts uploaded to cloud	Cloud-hosted	100% local & offline - nothing leaves the machine
Cost / internet	Paid API + internet	Infra + connectivity	Free, CPU-only, no internet
Track-02 (ML/DL) rule	LLM prediction = PROHIBITED	LLM-based = risky	Trained model = COMPLIANT

Bottom line: ChatGPT and cloud AI evaluators **generate** a score; ExamShield **predicts** it with a model **trained on your teachers' own marking** - explainable, measurable, private, and fully compliant with the Predictive Analytics track.

Evaluation metrics (full forms): **RMSE** = Root Mean Squared Error | **MAE** = Mean Absolute Error | **R^2** = R-squared (Coefficient of Determination) | **+/-1-mark accuracy** = percentage of answers whose predicted mark is within 1 mark of the teacher's mark.